

Chapter 133 — Use of Intestinal Segments in Urinary Diversion

- After extirpative surgery or for complex reconstruction, **bowel segments restore genitourinary function**; planning requires knowledge of anatomy and of the **physiological consequences of interposing gastrointestinal (GI) tissue** in the urinary tract.

SURGICAL ANATOMY

- **Ileum and colon are used most often**; stomach, jejunum and rectum less commonly. Success depends on knowing each segment's anatomy, vascular supply and metabolic function.

Stomach

- **Blood supply is mainly from the coeliac trunk**, through three branches:
 - **Left gastric (coronary) artery**: directly from the coeliac axis → lesser curvature.
 - **Hepatic artery** → **right gastric artery** (lesser curve) and **gastroduodenal artery** (antrum, duodenum) → **right gastroepiploic artery**.
 - **Splenic artery** → **vasa brevia (short gastrics)** to fundus and cardia, and the **left gastroepiploic artery**.
- **Right and left gastroepiploic arteries meet** and give collateral flow to the greater curve; on the gastroepiploic vessels a **gastric pedicle can reach the pelvis**. The pedicle is either the **entire antrum** or a **wedge of fundus**, based on the left or right gastroepiploic artery according to the portion used.
- **The left gastroepiploic artery is occasionally atretic**; the right must then be used.
- **A fundal wedge** must not include much antrum and must **never extend to the pylorus or to the lesser curve**.
- On a left gastroepiploic pedicle, the **short gastric vessels are ligated along the greater curve** proximal to the pedicle; the **omentum is left on the gastroepiploic vessels** to support them, and may be detached from the transverse colon along its avascular plane for mobility.
- **After antrectomy, a Billroth I anastomosis** restores GI continuity.
- The **thick seromuscular layer separates easily from mucosa**, which suits a submucosal ureteric reimplantation.

Small Bowel

- Length **about 22 feet** (range 15–30 feet); **proximal two-fifths jejunum, distal three-fifths ileum**, with no definite demarcation.
- Diameter is **largest in the duodenum** and **smallest in the ileum about 12 inches from the ileocaecal valve**.

JEJUNUM VERSUS ILEUM — INTRAOPERATIVE DISTINCTION

Feature	Jejunum	Ileum
Diameter	Larger	Smaller
Arterial arcades	Usually single	Multiple
Arcade vessels	Larger	Smaller
Mesentery	Thinner	Thicker

- Arcades (from the **superior mesenteric artery**) give **straight vessels** that form an intramural anastomotic network.
- Experimentally **up to 15 cm** of small bowel survives lateral to a straight vessel, but it is **unwise to assume more than 8 cm** will survive away from one.
- Transect the mesentery so the segment is fed by an **arcade supplied by a palpable artery of substance** in the base of the pedicle.
- **Two small-bowel regions lie in the pelvis** and are exposed to pelvic irradiation — avoid both in the irradiated patient:
 - the **last 2 inches of terminal ileum**, fixed in the pelvis by ligamentous attachments;
 - the **5 feet beginning about 6 feet from the ligament of Treitz**, which has the **longest mesentery** of the small bowel and so descends into the pelvis.

Colon

- Divisions: **caecum, ascending, transverse, left (descending), sigmoid colon and rectum**; some parts fixed or retroperitoneal, others free.
- **Caecum**: usually fixed in the right lower quadrant (rarely free and very mobile) by **two accessory peritoneal bands** — one from distal ileum via caecum to the retroperitoneum, one from caecum to the lateral posterior abdominal wall.
- **Ascending colon** fixed to the right posterior wall; at the hepatic flexure the **hepatocolic ligament** secures it to the liver.
- **Transverse colon** free; fixed at the splenic flexure by the **phrenocolic ligament**; attached to the stomach by the **gastrocolic omentum**.
- **Descending colon** fixed laterally; **sigmoid may or may not be free**; **rectosigmoid** intraperitoneal cephalad, then retroperitoneal, then subperitoneal.
- **Arterial supply: superior mesenteric, inferior mesenteric and internal iliac arteries**. Named vessels — ileocolic, right colic, middle colic, left colic, sigmoid, superior, middle and inferior haemorrhoidal — join to form the **marginal artery (arc) of Drummond**, which gives leeway in mobilisation.
- **Middle colic**: from the first part of the superior mesenteric artery (SMA), ascends the transverse mesocolon right of midline.
- **Right colic**: usually from the SMA just below the middle colic; variably from the ileocolic or middle colic.
 - If from the **ileocolic** → a distal ascending colon segment **reaches the deep pelvis more easily**.
 - If from the **middle colic** → the right colic may need **division at its origin** to bring distal ascending colon to the pelvis.
- **Ileocolic artery** = terminal SMA; supplies the **last 6 inches of ileum** and the ascending colon.
- Inferior mesenteric artery (IMA) → **left colic**, then **four to six sigmoid branches**; the last becomes the **superior haemorrhoidal artery**, which anastomoses with the **middle haemorrhoidal** (internal iliac), which anastomoses with the **inferior haemorrhoidal** (terminal internal pudendal). The **middle sacral artery** (from the aorta) may supply the posterior rectum.

- **Watershed areas** are fed by the most distal, small-calibre branches of two arteries, so are **vulnerable to ischaemia in hypotension or thromboembolism**.

COLONIC WATERSHED AREAS

Feature	Griffiths point	Sudeck's point
Site	Splenic flexure	Rectosigmoid junction
Communication	Middle colic ↔ ascending branch of left colic	Last sigmoid branch ↔ superior haemorrhoidal (rectal) artery
Significance	Perfusion of splenic flexure and descending colon	Tenuous; transection here may compromise the anastomosis

- Anastomoses at these points usually heal with good technique, but **site the anastomosis to one side of them**, where feeding arteries are larger.
- *Mobilisation sequence*. **Ascending colon**: divide caecal and distal ileal attachments → incise the **avascular line of Toldt** (bloodless if the mesentery is not breached). **Transverse colon**: divide gastrocolic omentum (avascular plane) → hepatocolic ligament (may carry small vessels) → phrenocolic ligament. **Descending colon**: incise the line of Toldt lateral to the colon. Further reach comes from **isolating a pedicle on a major named artery**.

KEY POINTS

- **Clear no more than 8 cm of mesentery** from the end of a small-bowel segment, to preserve perfusion and avoid necrosis.
- **Avoid anastomosis at the two colonic watershed areas** where possible: Griffiths point (splenic flexure) and Sudeck's point (rectosigmoid junction).

Concept — Every anatomical rule here is about **reach without ischaemia**: choose the pedicle that carries a named vessel, keep the segment's end within perfusion distance of a straight vessel, avoid watershed and irradiated zones, and exploit variant anatomy (right colic origin, gastroepiploic arcade) to gain length.

SELECTING THE SEGMENT OF INTESTINE

- **Selection depends on** the patient's condition, **renal function**, previous abdominal surgery, **mesenteric mobility** and the type of diversion or substitution required.
- **Neobladder**: success hinges on bowel mobility for a **tension-free, watertight urethral anastomosis**; body habitus and mesenteric length decide how far bowel reaches into the pelvis.
- **Manoeuvres to gain reach**: caecal mobilisation, **mesenteric Z-plasty**, use of a discarded segment at the bowel anastomosis, and **bed flexion** to reduce tension.

Stomach

- Used for **bladder replacement, augmentation cystoplasty, conduits and continent diversions**.
- **Advantages**: **less permeable** to urinary solutes; **net excretion of chloride and protons** (not absorption); **less mucus**. Urodynamically it behaves like other segments.
- **Electrolyte imbalance is rare with normal renal function**, though **hypochloraemic metabolic alkalosis** occurs. Earlier reports of less bacteriuria than ileum or colon are not borne out — **more recent data show no difference** between segments.
- Urine pH usually 6–7, so **no excess peristomal skin problems**; urinary pH differs little between gastric and ileal augmentation.

- **Serum gastrin usually normal or minimally raised**, depending on the portion and amount used; experimentally, **antral exclusion raises gastrin** and causes ulceration.
- **Severe ulcerative complications occur rarely** → consider **long-term histamine H2-receptor antagonists or proton-pump inhibitors (PPIs)**.
- **Do not use the antrum if the fundus is available**; if antrum is used, reconstruct with Billroth I, whose complications are well documented.
- **Early complications**: gastric retention (atony or anastomotic oedema); haemorrhage (usually anastomotic); hiccups from distension; pancreatitis from intraoperative injury; duodenal leak.
- **Delayed complications**: dumping, steatorrhoea, small-stomach syndrome, altered transit, bilious vomiting, afferent loop syndrome, hypoproteinaemia, megaloblastic or iron-deficiency anaemia.
- **Postoperative bowel obstruction** is appreciable in adults but rare in children; **gastroduodenal and gastroureteric leaks** occur, occasionally fatally.
- **Indications**: **severe renal dysfunction** (cannot tolerate metabolic acidosis); **risk of short-gut syndrome**; **dense abdominal adhesions** (the stomach is usually adhesion-free and easily mobilised).
- **Haematuria-dysuria syndrome** is specific to gastric segments and common: bladder spasm, suprapubic, penile or periurethral pain, gross haematuria or dysuria **without infection**, skin excoriation. Intermittent, self-limiting, **controlled with H2 blockers**. Attributed to **gastric acid**, though it occurs even with normal urine acidity.
- **Severe metabolic alkalosis with respiratory distress** occurs with gastric augments or conduits (see Metabolic Complications).

Jejunum

- **Usually avoided**: causes **severe electrolyte imbalance, most concerningly hyperkalaemia**.
- Diseases that preclude ileum usually preclude jejunum. If it is the only segment available, use **as distal a segment as possible**.

Ileum and Colon

- **Most often used**, in every type of reconstruction.
- **Ileum**: mobile, small diameter, **constant blood supply**; good for **ureteric replacement and conduits**.
 - Losing significant ileum → **vitamin B12 malabsorption, diarrhoea from bile-salt loss, fat malabsorption**.
 - Excess mesenteric fat or a **short mesentery** can hinder mobility or pelvic reach.
 - **Postoperative bowel obstruction up to 10%**; about half occur early.
- **Colon**: must be mobilised from fixed positions; **larger diameter**; reaches anywhere in abdomen or pelvis.
 - After pelvic irradiation, **right, transverse and descending colon are outside the field** and can be used confidently.
 - **Fewer nutritional problems than ileum** — provided the **ileocaecal valve is preserved**; using the valve → diarrhoea, bacterial overgrowth of ileum with malabsorption, fluid and bicarbonate loss.
 - **Bowel obstruction requiring reoperation 5%** — about **half the ileal rate**.
 - **Submucosal-tunnel antireflux ureteric anastomosis is easier** in colon.
- Both produce the **same electrolyte disturbance at similar frequency**; overall they are **comparable**, and neither is strongly preferred except in special circumstances.

SEGMENT SELECTION — COMPARISON

Segment	Electrolyte effect	Key advantage	Key disadvantage	Choose when
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Excretes H^+ and Cl^- ; less mucus; less permeable	Haematuria-dysuria syndrome ; ulceration; gastric-surgery complications	Renal failure; short gut; dense adhesions
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis	Outside radiation field	Severe electrolyte syndrome	Only when nothing else is suitable; most distal jejunum
Ileum	Hyperchloraemic metabolic acidosis	Mobile, constant blood supply	B12, bile-salt and fat loss; obstruction ~10%	Default conduit; ureteric replacement
Colon	Hyperchloraemic metabolic acidosis	Large, mobile; easy tunnelled antireflux; parts spared by radiation	Diarrhoea if ileocaecal valve used	Irradiated pelvis; antireflux anastomosis wanted

KEY POINTS

- **Stomach:** ulcerative complications and haematuria-dysuria syndrome, but **net excretion of H^+ and Cl^-** .
- **Jejunum:** severe electrolyte imbalance, **most concerning hyperkalaemia** — avoid if possible.
- **Excluding significant ileum** → vitamin B12 deficiency and reduced bile-salt reabsorption.
- **Colon** has a **lower rate of obstruction requiring reoperation** than ileum.

Concept — Segment choice is a trade between **what the bowel does to the urine** (stomach secretes acid, jejunum loses salt, ileum and colon absorb ammonium chloride) and **what its removal does to the gut**; the stomach and jejunum are reserved for patients in whom the standard ileum or colon would fail on one of those two counts.

PREPARING THE BOWEL FOR SURGERY AND POSTOPERATIVE RECOVERY

- **Complications from bacterial contamination** are a major cause of morbidity and mortality. Experimentally, an anastomosis with vascular compromise that would otherwise perforate **heals if the bowel has been antibiotic-prepared**; solid faeces strain an early anastomosis and can cause ischaemia and perforation.
- Faecal-contamination complications after radical cystectomy: wound infection, peritonitis, intra-abdominal abscess, wound and anastomotic dehiscence, systemic sepsis. Historically common, they have **fallen substantially with current practice**, though **infection of some kind remains frequent after cystectomy** — urinary infection most often, then superficial wound infection and bacteraemia.
- **Bacterial load:** stomach low; **jejunum, ileum and colon high**. Early studies linked unprepared bowel with more wound infection, abscess and dehiscence, and mechanically collapsed bowel with fewer leaks.
- **Meta-analyses of randomised trials in elective open colorectal surgery** do not support mechanical preparation for reducing leak or other complications. Small randomised studies of cystectomy with ileal conduit show **no difference in infection with or without preparation** — but they **excluded prior radiotherapy, immunosuppression, recent antibiotics, and liver or renal dysfunction**.
- **Normal flora:** aerobes — **Escherichia coli** and **Enterococcus faecalis**; anaerobes — **Bacteroides** and **Clostridium**.

MECHANICAL VERSUS ANTIBIOTIC PREPARATION

	Mechanical	Antibiotic
Acts on	Amount of faeces → total bacterial number	Bacterial concentration per gram of faeces
Concentration per gram	Unchanged	Reduced to about 10^2 organisms/g
Effect of a spill	Less to spill, but the same inoculum per millilitre	Smaller inoculum

FAECAL BACTERIAL CONCENTRATION BY SEGMENT (ORGANISMS PER GRAM)

Jejunum	Distal ileum	Ascending colon	Descending colon
10 to 10^5	10^5 to 10^7	10^6 to 10^8	10^{10} to 10^{12}

Mechanical Bowel Preparation

- **Reduces total bacteria, not concentration;** some analyses even suggest **more contamination after preparation**, perhaps from liquefied stool.
- **Conventional 3-day preparations** exhausted patients and worsened nutrition through low calorie intake.
- **Elemental diets failed:** they neither empty the colon nor reduce flora.
- **Whole-gut irrigation:** nasogastric tube (NGT) infusion of 9–12 L lactated Ringer's or saline over several hours; later **10% mannitol**, which cleaned equally well but **fed bacterial growth**.
 - Exhausting; may cause **fluid gain** (saline, mannitol).
 - **Contraindicated** in unstable cardiovascular status, cirrhosis, severe renal disease, congestive heart failure, bowel obstruction.
 - **No better than conventional preparation** for wound or septic complications, despite fewer aerobes.
- Largely replaced by **polyethylene glycol (PEG)–electrolyte solution**.
- **Evidence against:** limited preparations (half-volume PEG with metoclopramide; sodium phosphate) match full PEG; **two meta-analyses found more anastomotic dehiscence with mechanical preparation** (PEG possibly responsible); a meta-analysis of randomised trials of preparation versus none in elective colorectal surgery found **no difference** in leak, abscess or wound sepsis.
- **No adequate study for laparoscopic or robotic surgery.**
- **Caveats:** in these studies **intravenous antibiotics were crucial**; and colorectal resections involve limited bowel exposure, **unlike urological procedures**, where long segments are opened into a normally faeces-free urinary tract.

Antibiotic Bowel Preparation

- Oral antibiotic plus mechanical preparation dates from the **1970s**; early series showed **large reductions in septic complications, wound infection and mortality**. Antibiotics **protect vulnerable bowel**, letting a tenuous anastomosis survive.
- Contrary data: without oral antibiotics, septic rates can be comparable and **Clostridioides (Clostridium) difficile colitis is less frequent**.
- **Of little value in bowel obstruction** — they do not sterilise obstructed bowel.
- **Disadvantages:** postoperative diarrhoea; **pseudomembranous enterocolitis**; theoretical tumour implantation at the suture line (not relevant to urology); **candidal overgrowth** (stomatitis, thrush, diarrhoea); with prolonged use, protein, carbohydrate and fat malabsorption.
- **Agents:** kanamycin (**best single agent**); neomycin + erythromycin base; neomycin + metronidazole.

- **Recent evidence:** a randomised trial found the lowest faecal counts with mechanical preparation + oral neomycin + synbiotic, but no difference in clinical infection; a large colectomy series and a cohort of left-colon and rectal resections showed **fewer surgical site infections (SSIs), less C. difficile, fewer organ-space infections and leaks** when oral antibiotics were added to mechanical preparation (with intravenous antibiotics and chlorhexidine in both arms).
- **Perioperative intravenous antibiotics are the most important measure:**
 - must be **given before the operative event** — ideally **1–2 hours before incision**;
 - most effective against **anaerobes**, so the agent **must cover anaerobes**;
 - **third-generation cephalosporins** advocated; combined oral and intravenous prophylaxis supported; some stop oral agents for fear of C. difficile.
- *Authors' protocol.* Normal diet → **day before surgery: clear liquids + magnesium citrate + oral neomycin** → sports-drink hydration until 2 hours before surgery → **intravenous antibiotics 1 hour before incision**, per American Urological Association (AUA) guidance.

Wound Protectors

- **Physical barrier** between wound edges and viscera or instruments in clean-contaminated bowel surgery; limit bacterial inoculation and desiccation of wound edges.
- **Randomised trials and meta-analyses show fewer SSIs** in urological and colorectal procedures.

Enhanced Recovery After Surgery

- **Enhanced recovery after surgery (ERAS):** evidence-based perioperative pathways that **preserve organ function and blunt the surgical stress response** — counselling, nutritional optimisation, standardised analgesia and anaesthesia, early mobilisation. **Avoidance of routine bowel preparation** is a key component.
- Originated in **colorectal surgery in the 1990s (Henrik Kehlet)**; level 1 evidence there for **fewer complications and shorter stay**; since adopted across specialties, most recently urology.
- In urology, best studied in **radical cystectomy with diversion**. Bowel complications drive prolonged stay; **postoperative ileus (POI) is the commonest complication**.
- **Cystectomy patients are at higher POI risk than colectomy patients:** prolonged pelvic deperitonealisation during lymphadenectomy, bowel exposure to urine, longer operations, older and more comorbid patients, more smokers.
- POI definition varies; usually **prolonged intolerance of oral intake without mechanical obstruction**.
- Overall evidence supports ERAS in **well-selected cystectomy patients**; comparison is hampered by varied protocols and endpoints, and by cultural, expectation and reimbursement effects on stay.
- **Alvimopan: peripherally acting μ -opioid receptor antagonist;** blocks opioid binding in the gut, speeding upper and lower GI recovery; **limited blood-brain barrier penetration**, so central analgesia is preserved. Indicated after surgery including bowel resection; **inpatient use only** (US Food and Drug Administration, FDA).
 - **Dosing:** **12 mg 30 minutes to 5 hours before surgery**, then **twice daily from the day after surgery** until bowel function returns, **maximum 7 days (15 doses)**.
 - Value should persist whenever **opioids are still being given**, even within ERAS.
- ERAS has **not universally** shortened bowel recovery or stay, but in some series improved **opioid use, time to diet, intensive care unit (ICU) time and early quality of life**.
- **Component findings:** **chewing gum and metoclopramide** aid return of bowel function; **early enteral nutrition does not prevent postoperative protein depletion** (total protein, albumin, lymphocyte count).

TABLE 133.5 SUMMARY OF FINDINGS FROM PUBLISHED TRIALS PERTAINING TO THE USE OF ENHANCED RECOVERY AFTER SURGERY (ERAS) PROTOCOLS FOR RADICAL CYSTECTOMY

Evidence group	Length of stay	Complications / readmission / mortality	Bowel function and other
Controlled comparisons, open cystectomy (two studies)	Shorter with ERAS	No difference	Faster bowel movement, NGT and fluid removal, oral diet; shorter ICU stay
Controlled comparisons incl. robotic (two studies)	No difference (ICU time shorter in one)	No difference	Less opioid; earlier mobilisation and diet; better quality of life at discharge ; time to bowel movement unchanged
Uncontrolled ERAS series (four studies)	Short (most discharged within a few days in the best series)	Complications common; readmission and mortality low	Gum and metoclopramide help; early feeding does not prevent protein depletion
Alvimopan vs placebo — randomised, double-blind (Lee, 2014)	7.4 vs 10.1 days	Less POI ; readmission and mortality not different; cardiovascular events not increased	Bowel recovery 5.5 vs 6.8 days (text) / 5.5 vs 6.9 days (table)

Modified from Azhar RA, et al.: Eur Urol 70:176–187, 2016. Uncontrolled series had no control group.

- **What the table is for:** ERAS **does not increase complications or readmissions**; its benefit on stay is **inconsistent**, clearest in open series and in the alvimopan trial.
- **The alvimopan trial was not an ERAS trial:** patients received intravenous patient-controlled opioids without multimodal ERAS care. The two bowel-recovery values differ between the body text and the source table (**6.8 vs 6.9 days** for placebo), most likely a different data presentation in the systematic review from which the table is modified.
- **Trap:** "no difference in length of stay" in some trials does not mean ERAS failed — opioid use, diet, ICU time and quality of life still improved.
- **Prehabilitation:** before cystectomy, **nutritional** (high-protein diet), **functional** (tailored exercise) and **psychological** support, plus **smoking cessation** — aims to improve resilience, reduce complications and stay, and improve long-term recovery.
- **Caution outside cancer cystectomy:** ERAS trials excluded prior radiotherapy, immunosuppression, liver or renal dysfunction. **Benign-indication diversions** (spinal cord injury, chronic infection, radiation cystitis with damaged bowel, interstitial cystitis) carry poor motility, poor healing and impaired immunity.
- **Authors' practice.** **Radical cystectomy for malignancy → ERAS. Salvage cystectomy after trimodality therapy, or cystectomy for benign disease** (radiation cystitis, fistula to the pubic symphysis, neurogenic bladder) → **full mechanical and antibiotic preparation** and slow diet advancement.

KEY POINTS

- **Mechanical preparation** decreases faecal amount and so the **absolute bacterial count**.
- **Antibiotic preparation** decreases the **bacterial concentration** per millilitre of faeces.
- Whether to give mechanical and/or antibiotic preparation should weigh **risks and benefits for each patient**.
- **Alvimopan:** peripheral μ -opioid antagonist in the gut; limited blood–brain barrier crossing preserves central analgesia.

- ERAS evidence is largely **extrapolated from colorectal surgery**, but is supported for cystectomy and diversion for bladder cancer.
- **Very limited data** for ERAS in diversion for **benign conditions**.

Concept — *The evidence against bowel preparation comes from **colorectal resections in healthy, unirradiated patients under intravenous antibiotic cover**; urinary diversion opens long segments into a sterile tract, so the colorectal conclusion transfers only to the selected cancer-cystectomy patient, and **intravenous anaerobic cover before incision is the one non-negotiable**.*

INTESTINAL ANASTOMOSES

- In urological bowel surgery, **the commonest cause of early postoperative morbidity and mortality is bowel complications** — at the enteroenterostomy or the interposed segment.
- Bowel work often follows a long extirpation by a tired team, so the principles must be **ingrained, not recalled**.

Surgical Principles

- **1. Adequate exposure.** Mobilise so the anastomosis is done without struggle — ideally **on the anterior abdominal wall**; **wall off with Mikulicz pads** so spills are contained. **Clear mesentery about 0.5 cm** from clamps or staple line for serosal apposition without interposed mesentery.
- **2. Good blood supply to the cut ends.** Compromised by **tension, excess dissection, excess electrocautery, over-tight sutures**. A **pink, freely bleeding margin** indicates adequate supply (secure haemostasis before anastomosing).
 - Sequence: **transilluminate the mesentery** → choose a site where good arcades supply both ends, allowing for the depth of mesenteric division needed → clear mesentery from serosa → divide between mosquito clamps → tie with 4-0 silk (or LDS stapler, bipolar device, ultrasonic shears).
- **3. Prevent local spillage.** Best achieved with **prepared (faeces-free, collapsed) bowel**.
 - Sequence: **strip contents** away from the transection site → **non-crushing occlusive (linen-shod) clamp** that spares mesenteric flow → wall off → **Allen clamps** → transect between them (or transect with a stapler).
 - **Irrigate the isolated segment** with copious saline, walled off, into a kidney dish **until the efflux is clear**.
- **4. Accurate serosa-to-serosa apposition;** watertight, tension-free, gentle handling with non-crushing forceps. **Invert, do not evert:** everted anastomoses can heal, but under **marginal conditions an inverted anastomosis is more likely to hold**.
- **5. Do not strangulate tissue** with sutures. Non-absorbable sutures give a slightly stronger early anastomosis than absorbable ones, but the **difference is minimal**.
- **6. Realign the mesentery** parallel, with no twist. **Closing the mesenteric defect** is classically recommended but debated; bariatric data conflict on internal hernia, and there is little direct evidence in urinary diversion.
- **Factors causing anastomotic breakdown:** **poor blood supply, local sepsis from faecal spillage** (both via ischaemia), **drains placed on an intra-abdominal anastomosis, irradiated bowel**.

- In one series **most lethal postoperative complications of diversion were bowel-related**, and **most of those patients had been irradiated**; prior radiation significantly increases serious complications after radical cystectomy.

Concept — *All six principles serve two ends — **perfusion and a clean field** — because the named causes of breakdown (ischaemia, local sepsis, a drain on the line, radiation) each act by starving or contaminating the healing anastomosis.*

Complications of Intestinal Anastomoses

- Leak, sepsis, wound infection, abscess, haemorrhage, stenosis, **colonic pseudo-obstruction (Ogilvie syndrome)** and obstruction — major contributors to morbidity and mortality.
- In prepared bowel with contemporary technique, leak, haemorrhage and stenosis are **uncommon and rarely fatal**.
- **Leaks are diagnosed late** — typically in the **second postoperative week**, occasionally **after 30 days**.
- Many complications of radical cystectomy are **attributable to the bowel segment**.

Fistulas

- **Faecal or urinary**; usually within the **first several weeks**; frequently cause **sepsis**, which carries mortality.
- **Neobladder–vaginal fistula** after orthotopic diversion in women presents as **continuous incontinence several months after surgery**. **Vaginal-sparing cystectomy** reduces it by shortening opposing suture lines.

Sepsis and Other Infectious Complications

- **Wound complications are common**; dehiscence and pelvic abscess are rare. Septicaemia after cystectomy is uncommon but carries **substantial mortality**; cumulatively, **a large proportion of patients need treatment for infection** after diversion.
- **Prevention**: Mikulicz pads, irrigation of the segment until clear, avoid faecal spillage, **wash out abdomen and wound**.

Bowel Obstruction

- Obstruction requiring treatment **~10% with stomach or ileum**; **~5% requiring operation with colon** (Table 133.8). **Half occur early**; obstruction beyond 6 months is much less frequent.
- **Morbid**: a significant number of ileal-conduit patients needing operation for obstruction die.
- **Causes: adhesions (commonest), then recurrent cancer** — the great majority. **Volvulus and internal hernia are rare**. Anastomotic stenosis (oedema, poor technique, ischaemic bowel) or obstruction (improper technique) occasionally.
- **Prevention**: nonirradiated, well-vascularised bowel; **close all apertures**; **reperitonealise the segment**; adequate GI decompression; **omentum over the anastomosis**; **reconstitute the pelvic floor after exenteration**. **Orthotopic neobladders cause fewer obstructions** than conduits or cutaneous reservoirs.
- **Reperitonealisation**. Tack the **antimesenteric border to the lateral sidewall peritoneum** and the **proximal mesenteric border to the posterior parietal peritoneum** — an unobliterated space has trapped bowel.
- **Pelvic floor**. After **anterior exenteration**, lay the **sigmoid** in the pelvis, with **omentum** filling any residual space. After **total exenteration** (insufficient sigmoid, thin omentum) — especially if **postoperative pelvic radiotherapy** is planned — use **polyglactin mesh**: sutured posteriorly to the sacral promontory and presacral fascia, laterally to the iliac adventitia, anterolaterally to peritoneum **two-thirds of the way from pubis to umbilicus**, then covered with omentum. **Excludes bowel from the pelvis for 4–6 weeks of irradiation**.

TABLE 133.8 COMPLICATIONS OF URINARY INTESTINAL DIVERSION (INCIDENCE, %)

Complication	Ileal conduit	Colon conduit	Gastric conduit	Continent diversion
Bowel obstruction	10	4	10	0.8
Ureterointestinal obstruction	8	9; antireflux 20	—	4
Urine leak	3	5	—	Ileum 17 ; colon 4
Stomal stenosis, hernia	24	20	—	9
Renal calculi (pouch calculi for continent)	7	Antireflux 5	—	Pouch 13
Acidosis requiring treatment	16	Antireflux 5	0	Ileum 8; colon/colon-ileum 27
Pyelonephritis	12	Antireflux 13	—	5
Renal deterioration	18	Antireflux 15	—	—

Composite from the literature (multiple series). Mean follow-up: ileal conduits 5 years, colon conduits 3 years, gastric conduits 2 years, continent diversions 2 years.

- **What the table is for:** complication profile by diversion type. **Pattern:** antireflux colon anastomoses obstruct most; continent ileal reservoirs leak most; colon-based continent reservoirs cause most acidosis; gastric conduits cause none; stomal problems are the commonest conduit complication.
- **Text versus table** — the chapter's prose quotes different values for several endpoints, reflecting different source series and denominators:

Endpoint	Body text	Table 133.8
Colon conduit obstruction	5% (requiring operation)	4%
Antireflux colon pyelonephritis	9%	13%
Nonrefluxing colon renal deterioration	13%	15%
Continent ileal acidosis	10–15% (early reports)	8%
Ileal conduit renal calculi	10–12%	7%

- **Trap:** the lower acidosis figure for antireflux colon conduits is a **conduit** figure; **colonic continent reservoirs have the highest acidosis rate** — contact time and surface area matter more than the segment.

Haemorrhage

- **Rare;** far more likely with **stomach and a Billroth I**. Caused by **unsecured bleeding at anastomosis** or **anastomotic ulcers** on the suture line.

Intestinal Stenosis

INTESTINAL STENOSIS — EARLY VERSUS LATE

	Early (immediate postoperative)	Late (long term)
Cause	Technical mishap or oedema	Ischaemia or perienteric infection
Management	Oedema → continue decompression ; technical error → reoperate	Chronic partial obstruction

- **Figure 133.16.** Upper GI series: ileoileostomy stricture after ileal conduit; the suture line **looked bluish at operation**; at re-exploration a **2-mm lumen** with fibrotic serosa — **ischaemic stenosis**.

Pseudo-Obstruction

- **Ogilvie syndrome:** rare, cause unknown, usually **within 3 days** postoperatively in **patients with multiple medical illnesses**.

- Severe pain; radiograph shows a **dilated caecum**; a **gentle water-soluble contrast enema** excludes mechanical obstruction.
- **Rupture risk rises** with a rising leucocyte count or a **caecum enlarging beyond 12 cm**.
- Treatment: **neostigmine or endoscopic decompression**; surgery only for **ischaemia, peritonitis or refractory cases**.

Abdominal Stomas

FLUSH VERSUS PROTRUDING STOMA

	Flush	Protruding
Use	Continent diversion — intermittent catheterisation, small dressing	When a collection appliance is worn
Advantage	Low profile	Less stomal stenosis, better appliance fit, fewer skin problems
Types	—	End (nipple) stoma; loop end ileostomy

- **Most stomal complications are technical errors** — follow the technical points rigidly.
- **Siting (preoperative)**: mark **sitting and supine**; **over the rectus, at least 5 cm from the incision**; away from skin creases, scars, umbilicus, belt line, bony prominences and **irradiated skin**; at the **peak of the infraumbilical fat roll**.
- **Through the belly of the rectus** — **lateral to the rectus sheath** → **parastomal hernia**. Bowel exits **perpendicular** to the peritoneum. Do not trim fat or epiploic appendages at the margin; **apply the appliance in theatre**.
- **Aperture**. Finger-hole of a Kelly clamp on the site → Kocher on the central skin → cut **just beneath the Kocher** (not flush with the Kelly, which removes too much) → leave subcutaneous fat (it falls back and fills dead space) → Kocher on fascia, pulled medially so fascia and peritoneum are opened **directly under the skin opening** (no angulation) → **cruciate fascial incision** → spread rectus → open peritoneum → **admits two fingers snugly**.

Nipple (Rosebud) Stoma

- Sequence: Babcock brings bowel out **5–6 cm** for a **2–3 cm nipple** → two 3-0 chromic seromuscular-to-peritoneum sutures (or serosa to fascia with 2-0 chromic), mesentery aligned → if the ileum curves severely toward its mesentery, **incise the mesentery 1 cm from the bowel wall** to straighten it (Figure 133.19) → **four quadrant 3-0 chromic sutures**: full-thickness bowel edge → seromuscular layer **3–4 cm** proximal → **subcuticular/subdermal skin only** (never full-thickness skin) → tie to evert.
- **Seromuscular myotomies** above skin level before eversion give serosa-to-serosa adhesion and **reduce retraction** — useful in **obesity**.

Flush Stoma

- Quadrant 3-0 chromic sutures, full-thickness bowel to subdermal skin, with intervening sutures → stoma with a **1-mm raised margin**.

Loop End Ileostomy

- For **obese patients** (thick wall, thick short mesentery) — **easier than an end stoma**; keeps the mesenteric border off the abdominal wall.
- Sequence: close distal end → **3-cm skin disc**, opening admits **two fingers comfortably** → umbilical tape through a mesenteric window draws the loop through the rectus, **closed end cephalad** → **rod** through the mesentery holds it → close an oversized rectus rent from within (0 chromic) → serosa to peritoneum → **transverse enterotomy four-fifths of the way cephalad** → caudal edge everted (rosebud fashion, 3-0

chromic to dermis); **cephalad non-functional opening sutured flat to dermis → rod sutured to skin for 7 days.**

END VERSUS LOOP (TURNBULL) STOMA

	End stoma	Loop (Turnbull) stoma
Stomal stenosis	Higher	Lower
Parastomal hernia	Rare (1–4%)	Higher (4–20%)
Best for	Standard patient	Obese patient
Long term	Functionally equivalent	

- **Colonic stomas** are made like ileal end stomas but sutured **more flush than everted**.

Complications of Intestinal Stomas

- **The single most common postoperative problem** after diversion; at some point **virtually every patient** has one.
- **Early:** bowel necrosis, bleeding, dermatitis, parastomal hernia, prolapse, obstruction, retraction, stenosis.
- **Reduced by:** correct construction; **enterostomal therapist** review; proper skin products and non-irritant adhesives; **keeping urine in the bag acidic**; well-fitting appliances.
- **Parastomal skin lesions:** **irritative** (hypo-/hyperpigmentation, atrophy); **erythematous erosive** (macules, scaling, epidermal loss); **pseudoverrucous** (wart-like).
- **Stomal stenosis:** about **20–24% of ileal conduits** and **10–20% of colon conduits**; falling with better care and appliances; **less with loop than end stomas**. **Catheterisable stomas in children** stenose very frequently (over half).
- **Umbilical stomas:** some report more complications, others excellent results with the concealed umbilical stoma.
- **Bleeding, stenosis and dermatitis** respond to skin care and a well-fitted appliance on a protruding stoma; the rest are prevented by technique.
- **Parastomal hernia after ileal conduit:** **2–6.6%**; **open repair effective**; laparoscopic repair has mixed results, including high early failure with expanded polytetrafluoroethylene mesh in one series.
- **Massive conduit bleeding:** usually **stomal/conduit varices** from **hepatic dysfunction and portal hypertension** → **transhepatic portosystemic shunt, or transhepatic angiography with embolisation**.

KEY POINTS

- Anastomotic principles: **(1) exposure, (2) blood supply, (3) minimal spillage, (4) serosa to serosa, (5) no strangulation, (6) mesenteric alignment**.
- **Sutured and stapled anastomoses** have similar complication rates when properly performed.
- The **stapled side-to-side ileoileal anastomosis** is the most commonly performed, for simplicity and reliability.
- **Stomal complications** are mostly technical and are the **most common problem** after diversion.
- **All stomas through the belly of the rectus** to minimise parastomal hernia.

Concept — *Stoma design is an exchange between **stenosis and hernia**: protrusion and the loop configuration protect against stenosis and skin injury, but the wider loop aperture invites herniation — which is why the loop is kept for the obese patient in whom an end stoma cannot be built.*

URETEROINTESTINAL ANASTOMOSES

- The ureter joins colon or small bowel in a **refluxing or nonrefluxing** fashion; which is preferable is **controversial**.
- Over the long term, **renal failure affects a meaningful minority** regardless of diversion type, and **upper-tract deterioration after conduits is reported in 10–60%**.
- **Causes of upper-tract deterioration: lack of ureteric motility, infection, stones; less commonly** obstruction at the anastomosis.
- Because **bacteriuria is almost universal** in conduits and intestine may promote colonisation, many argued that **nonrefluxing anastomoses** would protect the kidney.

EVIDENCE ON REFLUX AND RENAL DETERIORATION

For nonrefluxing anastomosis	Against (reflux not harmful per se)
• Nonrefluxing colon conduits that stayed nonrefluxing deteriorated far less than those whose antireflux mechanism failed (long follow-up). • Continent diversions with reflux show upper-tract dilatation and deterioration. • Experimentally, refluxing ureterocolonic conduits scar far more than nonrefluxing ones, despite similar bacteriuria.	• Colon conduit series with equal deterioration whether or not reflux was present. • Unobstructed refluxing anastomoses rarely deteriorated. • Comparative conduit study: no difference in deterioration or pyelonephritis, but stricture far more frequent with nonrefluxing units. • High-capacity, low-pressure reservoirs and orthotopic substitutes do not justify the stricture risk of antireflux techniques.

- **Conduit pressure is not transmitted to the renal pelvis** (ileum or colon): **ureteric peristalsis damps transmission** — hence the importance of normal ureters. Where bowel replaces the ureter, reflux at the bladder is immaterial; the distensible segment blunts voiding pressure.
- Ileal and colon conduits show **no difference in renal function at 2–5 years** between refluxing and nonrefluxing units.
- **Antireflux does not prevent renal pelvic colonisation** (positive pelvic aspirates despite documented absence of reflux).
- **Advantage of reflux in urothelial cancer:** the upper tracts can be **imaged by instilling contrast into the conduit**.
- **Conclusion:** reflux + impaired peristalsis + bacteriuria or obstruction → renal deterioration; reflux with a normal, unobstructed ureter has not been shown to harm the adult kidney.
- **General principles (all types):**
 - Mobilise **only as much ureter as needed** — no redundancy, no tension.
 - **Preserve periadventitial tissue** (it carries the blood supply); clear adventitia only for the **distal 2–3 mm**.
 - **Fine absorbable sutures;** watertight **mucosa-to-mucosa** apposition.
 - **Bring the bowel to the ureter,** not the ureter to the bowel.
 - **Fix the bowel** near the anastomosis; **retroperitonealise** it or cover with a peritoneal pedicle flap.
- **Fixation sites** that spare mesenteric flow: **below the root of the small-bowel mesentery at the pelvic brim** (most convenient); or, for **short ureters,** the **right upper quadrant cephalad to the right colic takeoff, just below the duodenum** — relatively avascular and close to both kidneys.

- **Strictures:** from **ischaemia, urine leak, radiation or infection**. **Urine leak 3–5% overall, reduced nearly to zero with soft Silastic stents.**
 - Evidence: **rigid non-Silastic stents increased stricture**; soft Silastic stents abolished leak and stricture in one institutional series; stenting reduced leak and stricture after gynaecological exenteration; a **randomised study showed fewer early metabolic complications with stents.**
 - **Modern soft Silastic stents reduce leak, stricture and postoperative complications**; better technique and sutures have also lowered stricture after nonrefluxing anastomoses.
- **Submucosal tunnel tip: inject saline through a 25-gauge needle** to lift mucosa off the seromuscular layer.

Ureterocolonic Anastomoses

Combined Technique of Leadbetter and Clarke

- **Nonrefluxing, submucosal tunnel:** combines **Nesbit's refluxing elliptical anastomosis** with **Coffey's tunnel.**
- Sequence: incise anterior **taenia obliquely 2.5–3 cm**, near the mesenteric border → dissect mucosa off muscularis for the incision's length → excise a **mucosal button at the distal end** → **spatulate ureter 5–7 mm** → **mucosa to mucosa with 5-0 polydioxanone (PDS)**, interrupted (knots outside) or running from the apex → **loosely close seromuscular layer** so the ureter lies "as a hammock" uncompressed → fix bowel to peritoneum, no ureteric tension.
- Complications: **leak 2.5%; upper-tract deterioration 4.3–25%; stricture 8–14%.**

Transcolonic Technique of Goodwin

- **Nonrefluxing submucosal tunnel, performed from inside the bowel** (open colotomy).
- Sequence: loose non-crushing clamp cephalad (bowel in continuity) → **anterior vertical colotomy** → **0.5-cm posterior mucosal incision** → hemostat tunnels submucosally, medial to lateral, **3–4 cm** → out through serosa → draw ureter in by traction suture → **bring both ureters in before suturing either** → pass a **No. 5 feeding tube** to exclude kinking → excise redundant ureter, spatulate, **interrupted 5-0 PDS to mucosa including some muscularis** → **Silastic stents** → ureteric adventitia to colonic serosa at entry (**4-0 PDS** in the text; interrupted 5-0 PDS in the figure legend) → close colotomy in two layers.
- Results satisfactory; **no reliable complication data.**

Small Bowel Anastomoses

- Two basic types: **end to side** (refluxing or nonrefluxing) and **end to end.**

Bricker Anastomosis

- **Refluxing end-to-side; simple, low complication rate**; usable in any suitable segment.
- Original: silk adventitia-to-serosa posterior layer → mucosal plug removed → full-thickness ureter to mucosa (chromic) → anterior silk layer.
- **Modern:** excise a small full-thickness button → **spatulate 0.5 cm** → full-thickness ureter to full-thickness bowel with **5-0 PDS** → **soft Silastic stent.**
- **Stricture 4–22% (average 6%); leak ~3% without stents; open and robotic stricture rates comparable.**

Wallace Technique

- **Refluxing end-to-end:** the ureters are joined and sutured to the **end of the bowel** (small bowel or colon).
- **Three variants:**
 - **Side-by-side:** apices joined, **posteromedial walls** sutured together, **anterolateral walls sutured to bowel.**

- **Y-type:** posterior then anterior walls sutured together; the composite end sewn to bowel.
- **Head-to-tail:** end of one ureter to the apex of the other; posteromedial walls joined; anterolateral walls to bowel.
- **Spatulate 1.5–2 cm; 5-0 PDS; horizontal mattress where suture lines meet the bowel** for watertightness.
- **Lowest complication rate of any ureterointestinal technique:** stricture ~3%, deterioration ~4%, leak ~2%.
- Historically **avoided in extensive carcinoma in situ (CIS)** or likely ureteric recurrence (one tumour could **block both ureters**) — but this is **exceedingly rare**.
- **Easier endoscopic surveillance** (both orifices side by side at the proximal loop) and **lower obstruction rate than Bricker**.

URETEROINTESTINAL ANASTOMOSES — COMPARISON

Technique	Reflux	Configuration	Spatulation	Stricture	Leak
Leadbetter-Clarke	Nonrefluxing	Colon; extraluminal submucosal tunnel via taenia	5–7 mm	8–14%	2.5%
Goodwin	Nonrefluxing	Colon; tunnel from inside the bowel	—	No reliable data	—
Bricker	Refluxing	End to side; any segment	0.5 cm	4–22% (mean 6%)	~3% unstented
Wallace	Refluxing	End to end, conjoined ureters	1.5–2 cm	~3% (lowest)	~2%

- **Pattern:** the tunnelled (antireflux) techniques stricture more; the refluxing conjoined Wallace does best.

Complications of Ureterointestinal Anastomoses

- **Leak, stricture, reflux** (in antireflux anastomoses) and **pyelonephritis**.
- **Best antireflux options:** colon — **Pagano technique** (lowest stricture, acceptable reflux); small bowel — **Le Duc** and **ureterointestinal serosal apposition** (lowest stricture, best reflux prevention). For stricture and leak overall, **Wallace is best**.

Urinary Fistula

- Usually **days 7–10; 3–9%**; **markedly reduced by soft Silastic stents**. A leak causes **periureteric fibrosis** → **later stricture**.

Stricture

- **Antirefluxing techniques stricture more**. Risk is **lifelong** (reported at 13 years) → **scheduled lifelong surveillance**.
- **Strictures away from the anastomosis:** typically the **left ureter where it crosses the aorta beneath the IMA** — from **over-stripping of adventitia and angulation** at the IMA.
- **Treatment:** **open re-exploration, excision and reanastomosis is most successful** and durable, though morbid; **balloon dilatation is not durable**; **cold-knife or laser endoincision** succeeds in about half to three-fifths at 2 years (worse with longer follow-up).
- **Poor endourological outcome:** stricture **within 1 year** of surgery, **≥1.5 cm long**, or **left-sided**.
- **Metal stents:** for **limited life expectancy**; use with caution — a metallic ureteric stent series showed poor early patency; **self-expanding metal alloy stents** did better in small benign series (one stent encrustation). Long-term data needed.

Pyelonephritis

- Early and late; about **10–20% after ileal conduits** and lower after antirefluxing colon conduits (Table 133.8 gives similar rates for both).

- **Significant mortality**, including **delayed septic deaths years after diversion** in children and adults; **sepsis with falling renal function and uraemia** markedly raises morbidity and mortality. **Multidrug-resistant organisms** are a growing threat.
- Rates across anastomotic types cannot be compared inclusively for reflux or leak; complications are minimised by the principles above.

KEY POINTS

- Complications: **leak, stricture, reflux** (in antireflux anastomoses), **pyelonephritis**.
- **Stents markedly reduce** urinary intestinal leak.
- **Antirefluxing techniques stricture more** than refluxing ones.
- **Wallace has the lowest complication rate**.
- Long term, **most patients show progressive renal deterioration regardless of diversion type** — from poor ureteric motility, infection, stones, or less commonly anastomotic stricture.

Concept — *The kidney is protected by **normal ureteric peristalsis and freedom from obstruction**, not by an antireflux valve; because every tunnel adds stricture risk, a well-made **refluxing, stented, tension-free anastomosis** is the default and the antireflux tunnel the exception.*

RENAL DETERIORATION

- Reported after conduit diversion in **10–74%** — wide because series **mix previously normal and abnormal renal units**; in abnormal units, diversion and underlying disease cannot be separated.
- In **previously normal units: ileal conduit 18%** versus **nonrefluxing colon conduit ~13–15%** (Table 133.8); nonrefluxing continent ileocaecal reservoirs also deteriorate in a minority long term. Consequences include **azotaemia in children with ileal conduits** and **renal failure after colon conduits for benign disease**.
- **Significant, progressive decline in most patients, independent of diversion type**.
- **Ureterosigmoidostomy carries more sepsis and renal failure than conduits**, early or many years later. **After more than 15 years, acquired renal disease (sepsis or renal failure) is the commonest cause of death** (10–22% die of it), some as late as 27 years. About **6% of ileal conduit patients** ultimately die of renal failure.

Renal Function Necessary for Urinary Intestinal Diversion

- Required function depends on **how much bowel** and **how long urine dwells** → **continent (retentive) diversions need more renal function than short conduits**.
- **GFR (glomerular filtration rate) above 40 mL/min on average** → continent diversion tolerated reasonably well.
- **GFR is not the sole determinant — distal tubular function is equally important**: patients with moderately reduced GFR but intact acidification on ammonium chloride loading tolerated continent diversion with little acidosis.
- Five components of renal function: **renal blood flow, glomerular filtration, tubular transport, concentration and dilution, glomerular permeability**.

ASSESSING RENAL FUNCTION BEFORE DIVERSION

Parameter	Best test	Threshold for a retentive diversion
Screening	Serum creatinine + urine protein	Creatinine <2.0 mg/dL with normal urine protein → proceed
GFR	Inulin clearance	>35 mL/min
Tubular acidification	Ammonium chloride loading	Urine pH ≤5.8
Concentrating ability	Water deprivation	Urine osmolality ≥600 mOsm/kg
Glomerular permeability	Urine protein	Minimal proteinuria

- Below creatinine 2 mg/dL, blood flow, GFR, tubular transport and concentrating ability are relatively preserved. **Above 2 mg/dL**, if a retentive diversion or long segment is planned → **full detailed testing**.
- **Serum cystatin C** reflects GFR better than creatinine, but its role in diversion is **not yet established**.
- **Exception:** retentive diversion can be built in **end-stage renal failure**, particularly in **children awaiting transplantation**, with close nephrology collaboration (see Chapter 61).
- **Trap:** GFR >40 mL/min is the average tolerance figure; the formal work-up threshold is **>35 mL/min** — together with pH and osmolality criteria.

Concept — *The kidney must **excrete what the bowel reabsorbs**; so the more bowel and the longer the dwell, the more renal reserve — above all **tubular acidifying capacity** — the patient needs.*

URINARY DIVERSION

- Two fundamental conduit types: **small bowel (jejunum or ileum)** and **large bowel**. **Gastric conduits** are rarely indicated and cause **difficult stomal problems**; not described.
- **Indications for a conduit:** after **cystectomy**; a **diseased bladder**; **before transplantation** when the bladder cannot receive the transplant ureter; **dysfunctional bladders** causing persistent bleeding, ureteric obstruction, **poor compliance with upper-tract deterioration**, or inadequate storage with total incontinence.
- Complication rates depend on follow-up length and the concomitant procedure. Orthotopic neobladders are covered in a separate chapter.

Ileal Conduit

- **Simplest conduit**, with the **fewest intraoperative and early postoperative complications**.
- **Avoid ileum** in **short bowel syndrome**, **inflammatory small-bowel disease**, and **extensively irradiated ileum** (usually after pelvic radiotherapy).

Procedure

- Sequence: select a **10–15 cm segment 10–15 cm from the ileocaecal valve** → mobilise caecum and ileal appendage (distal ileum fixed to retroperitoneum) → **transilluminate**, identify a major arcade → vessel loops beneath the bowel and at the mesenteric base to one side of the feeding vessel → incise peritoneum on both sides of the mesentery between loops → clamp, divide, tie with 4-0 silk → **clear 2 cm of mesentery** from bowel → repeat at the other end → **wide mesenteric base; windows about 5 cm**, not excessive (ischaemia) → Allen clamps angled so the **antimesenteric side is shorter** (triangle of bowel discarded), or staple.
- Segment placed **caudal to the ileoileostomy** (Figure 133.34) → close the mesenteric window with 3-0 silk → **flush segment until clear**.
- **Ureters.** Brought out of the retroperitoneum in the **right lower quadrant**; the **left ureter passes over the great vessels, behind the sigmoid mesentery**. Caecum mobilised cephalad to find the right ureter; **line of**

Toldt incised to find the left (Figure 133.35) — allows anastomosis as proximal as needed, even **directly to the renal pelvis**. After cystectomy, trace the ureters up from below the iliac vessels.

- **Stent the anastomoses** — conveniently via a **Yankauer sucker with its tip removed**, passed from the distal end and cut down upon at the anastomotic site (Figure 133.36).
- **Fix the conduit base to the retroperitoneum** in the right lower quadrant, retroperitonealising the anastomosis → stoma as described.
- **Authors suture the loop to the lateral peritoneal wall** (no lateral internal hernia); many bring it straight to the anterior wall, letting bowel descend on either side.
- **Robotic and laparoscopic ileal conduits: similar 90-day complication rates** in experienced hands.

Complications

- Hard to attribute solely to the conduit (most series include cystectomy) — published rates represent the **high end**.
- **Bleeding** from stoma or conduit: stomal bleeding in about **10%**, a minority from the loop beneath the fascia. **Intractable bleeding → suspect cirrhosis with varices**; may be life-threatening → **portal decompression** (percutaneous transhepatic portal shunt, or transhepatic angiography with embolisation).
- Hypertension, renal failure, declining function and death depend on the concomitant procedure, follow-up and pre-diversion renal status. **Over 20 years, a minority need dialysis but most show upper-tract deterioration**.
- **Salvage cystectomy**: acceptable morbidity but **higher complication rates** than other cystectomy series.
- **Higher complications** when an intestinal segment is used in patients who later need **renal transplantation**.
- **Duplex ureters**: missing a second ureter → **intraperitoneal urine leak** and excessive morbidity. Implant separately if calibre allows, or spatulate, sew together and implant as one (Figure 133.37).

SALVAGE CYSTECTOMY AFTER CHEMORADIATION — IMMEDIATE VERSUS DELAYED

	Immediate salvage	Delayed salvage
Setting	Persistent disease after induction chemoradiation	Invasive recurrence after completed chemoradiation
Excess complications	Cardiovascular / haematological	Tissue healing

Jejunal Conduit

- Jejunum has the **largest small-bowel diameter and longest mesentery**. Little used because of **electrolyte abnormalities**, though long-term data suggest these are **mostly minor**, with only a small minority developing **severe hyponatraemic metabolic acidosis**; renal calculi, parastomal hernia and pyelonephritis make up most other complications.
- **Advantage: avoids irradiated bowel and ureter**. Justified only when **ileum and colon are both unsuitable**: extensive ileal irradiation; dense ileal adhesions with absent large bowel; absent colon with inflammatory disease of distal small bowel.
- **Contraindications: severe bowel nutritional disorders; another acceptable segment available**.

Procedure

- As for ileal conduit: **10–15 cm segment isolated 15–25 cm from the ligament of Treitz**; plan the **stoma in the left upper quadrant**.

Complications

- As for ileal conduit, except the metabolic disturbance is **hyperkalaemic, hyponatraemic metabolic acidosis** rather than hyperchloraemic acidosis.
- **Jejunal syndrome treatment: sodium chloride and sodium bicarbonate; thiazide diuretics** help control hyperkalaemia.

Colon Conduit

COLON CONDUITS — COMPARISON

	Transverse	Sigmoid	Ileocaecal
Best indication	Extensive pelvic irradiation (segment certainly unirradiated); intestinal pyelostomy	Pelvic exenteration with colostomy — no bowel anastomosis needed	Long ileum to replace long ureteric segments + colon for the stoma; when reflux is undesirable
Other advantages	—	Nonrefluxing submucosal reimplantation; easy left-sided stoma	Ileocaecal valve can be reinforced against reflux
Specific contraindications	—	Sigmoid disease; hypogastric arteries ligated with rectum left in situ (rectal slough); extensive pelvic irradiation	—
Blood supply	Middle colic (commonest) or right colic	Sigmoid vessels	Ileocolic (terminal SMA)
Stoma	Usually right upper quadrant	Left	Right lower quadrant

- **Contraindications to all colon conduits: inflammatory large-bowel disease and severe chronic diarrhoea.**

Procedure

- **Transverse colon.** Take down the gastrocolic ligament, strip omentum from the segment → mobilise splenic and hepatic flexures → length set by stoma site and ureteric length, **usually 15 cm; never too short** to reach the retroperitoneum for a tension-free, retroperitonealised anastomosis → isolate; two-layer or stapled colocolostomy → segment **caudal** to the anastomosis (**cephalad for colopyelostomy**) → irrigate until clear → close proximal end: **running 3-0 chromic Connell + 3-0 silk Lembert** → ureterocolic anastomoses → anchor end to retroperitoneum near midline.
- **Sigmoid colon.** Incise peritoneal attachments and line of Toldt → isolate on sigmoid vessels → place **lateral to the sigmoid** (Figure 133.39) → anastomoses as for transverse.
- **Ileocaecal.** Place caudad → ileum-ascending colon anastomosis → stoma right lower quadrant → optionally **reinforce the ileocaecal valve** against reflux.

Complications

- Death, renal failure and deterioration again depend on the concomitant procedure and follow-up. Early reports suggested **less renal deterioration with colon conduits**; recent series suggest **about the same** as ileal conduits, though long-term pyelonephritis is low and most upper tracts are preserved.
- **Ileocaecal versus ileal conduit: no difference** in early or late complications or in upper-tract deterioration. Early: leak, obstruction, faecal leak, renal failure, fulminant hepatitis, pneumonia, GI bleeding, ileal perforation, heart failure, dehiscence. Late: stomal prolapse and stenosis, pyelonephritis, obstruction, stones, parastomal and incisional hernia, faecal leak.
- **Ileocaecal conduits reflux at high pressure**, minimally at low pressure.

- **Any colon conduit may cause chronic diarrhoea.**

Ileal Vesicostomy

- **Spatulated ileum on a generous transverse cystotomy**, with a distal **rosebud stoma** → **low-pressure reservoir** draining to an abdominal appliance.
- For **spinal cord injury or significant neurological disease**; best candidates have **detrusor-external sphincter dyssynergia**. **Detrusor hyperreflexia (neurogenic detrusor overactivity)**, particularly in women, raises the risk of **urethral incontinence**.
- Complications: **urethral incontinence requiring urethral closure** in a significant minority; stomal stenosis; bladder and renal calculi.
- **Reversible** — anatomy can later be restored.

Management Common to All Conduits

- **All anastomoses stented with Silastic stents**, removed individually on **postoperative days 4–6**; if drainage does not rise, remove the **Jackson-Pratt closed-suction drain**.
- **All conduits are retroperitonealised**: posterior peritoneum sutured to conduit serosa above the anastomosis.
- **Drain** (Jackson-Pratt or Blake) in the retroperitoneum **3–4 cm from the anastomosis**; **never drain the peritoneal cavity**.

KEY POINTS

- **Salvage cystectomy**: acceptable morbidity, higher complication rates.
- **Ileal conduit**: simplest, fewest intraoperative and early complications.
- **Transverse colon**: least likely colonic segment to have been irradiated.
- **Sigmoid conduit**: good choice at pelvic exenteration with colostomy — no bowel anastomosis.

Concept — Conduit choice follows **the radiation field and the bowel already being sacrificed**: ileum by default, transverse colon when the pelvis is irradiated, sigmoid when a colostomy is being made anyway, jejunum only when nothing else remains.

METABOLIC AND NEUROMECHANICAL PROBLEMS OF URINARY INTESTINAL DIVERSION

- Three categories: **metabolic** (altered solute reabsorption from the contained urine), **neuromechanical** (gut configuration affecting storage volume and contraction), and **technical-surgical** (covered with each procedure).

Metabolic Complications

- **Spectrum**: electrolyte abnormalities, altered sensorium, abnormal drug metabolism, osteomalacia, growth retardation, persistent and recurrent infection, renal and reservoir calculi, consequences of removing gut, and **urothelial or intestinal cancer**.
- **Determinants of solute absorption**: segment used, surface area, urine dwell time, urinary solute concentration, renal function, pH.

Electrolyte Abnormalities

TABLE 133.15 SYNDROMES OF ELECTROLYTE DISTURBANCES IN PATIENTS IN WHOM BOWEL IS INTERPOSED IN THE URINARY TRACT

Segment	Syndrome	Na ⁺	K ⁺	Cl ⁻	pH	Associated	Symptoms	Treatment
Stomach	Severe metabolic alkalosis	—	↓	↓	↑	Raised aldosterone	Lethargy, weakness, respiratory insufficiency, seizures, ventricular arrhythmia	H2 blocker, PPI; if life-threatening, arginine hydrochloride and/or segment removal
Jejunum	Hyperkalaemic, hypochloaemic metabolic acidosis	↓	↑	↓	↓	Raised renin and angiotensin	Lethargy, nausea, vomiting, dehydration, weakness	Intravenous hydration, sodium bicarbonate, thiazide; if life-threatening, segment removal
Ileum / colon	Hyperchloaemic metabolic acidosis	—	↓	↑	↓	Total-body potassium depletion; hypocalcaemia	Fatigue, anorexia, lethargy, weakness	Potassium citrate, sodium citrate, citric acid, sodium bicarbonate, chlorpromazine, nicotinic acid

- **What the table tests: chloride moves opposite to the syndrome's name** — stomach secretes HCl (alkalosis, low Cl⁻); jejunum secretes NaCl (low Na⁺ and Cl⁻, high K⁺); ileum and colon absorb NH₄Cl (high Cl⁻). **Trap: potassium is low in both stomach and ileum/colon** — only jejunum is hyperkalaemic.
- Other reported abnormalities: **hypokalaemia, hypomagnesaemia, hypocalcaemia, hyperammonaemia, raised urea and creatinine.**

Stomach.

- **Hypochloaemic, hypokalaemic metabolic alkalosis** — usually trivial unless **renal failure** (impaired bicarbonate excretion) or **marked dehydration**.
- **Syndrome of severe metabolic alkalosis** (occurs even with normal renal function): lethargy, respiratory insufficiency, seizures, ventricular arrhythmias; usually **preceded by vomiting and dehydration**.
- **Treatment ladder: H2 blocker + rehydration** → life-threatening: **arginine hydrochloride infusion** → H2 failure: **omeprazole** → persistent: **remove the gastric segment**.
- **Gastrin is pivotal**: serum gastrin correlates with bicarbonate in gastrocystoplasty patients. *Vicious circle*: gastric proton loss adds bicarbonate → vomiting causes volume depletion, hypochloaemia, hypokalaemia → **aldosterone rises** → the kidney **cannot excrete bicarbonate** → alkalosis worsens.
- **Sigmoid gastrin-bicarbonate curve** (Figure 133.40): over normal gastrin **10–120 ng/L**, bicarbonate barely changes; **above 120 ng/L**, small gastrin rises cause large bicarbonate rises; **below 100 ng/L**, gastrin can rise considerably with little effect.
- **Distension stimulates gastrin**, so failure to empty raises risk. Patients with **resting gastrin >120 ng/L** or **impaired renal function** must be warned about **dehydration and overdistension**.

Jejunum.

- Especially with **proximal jejunum**: hyponatraemia, hypochloaemia, hyperkalaemia, azotaemia, acidosis.
- **Mechanism (cycle)**: jejunum **secretes NaCl** (with water) and **reabsorbs K⁺ and H⁺** → dehydration, hypovolaemia → **renin-aldosterone rises** (also driven by hyperkalaemia) → kidney retains Na⁺, excretes

K^+ → **low-sodium, high-potassium urine** reaches the jejunum → steeper gradient for Na^+ loss and K^+ gain → perpetuation.

- Symptoms: lethargy, nausea, vomiting, dehydration, weakness, **fever**; untreated → moribund → death. **Worsened by parenteral nutrition** (mechanism unclear). **More proximal segment → more likely**; incidence ranges from a quarter to most patients, but severe forms are uncommon with short segments.
- **Treatment: rehydrate with sodium chloride, correct acidosis with sodium bicarbonate**; hyperkalaemia corrects via the kidney if renal function is normal (diuretic occasionally). Long term: **oral sodium chloride; thiazide** for hyperkalaemia.

Ileum and colon.

- **Hyperchloraemic metabolic acidosis** occurs to some degree in **most patients**, usually minor and of unclear significance; a minority need treatment and few need hospitalisation. Severe disturbance is **far commoner with ureterosigmoidostomy** (most patients) than with conduits or cystoplasties.
- **Continent ileocaecal or caecal reservoirs**: most have raised chloride and low bicarbonate; **most Mainz pouch patients need alkali**. Early reports of ileal continent diversions showed far fewer problems.
- Severe symptoms: **fatigue, anorexia, weight loss, polydipsia, lethargy**; ureterosigmoidostomy also worsens diarrhoea. Severe disturbance **can be fatal**.
- **Mechanism: ionised ammonium transport** — NH_4^+ substitutes for Na^+ on the Na^+/H^+ antiport, coupled with Cl^-/HCO_3^- exchange → **ammonium chloride absorbed** in exchange for carbonic acid (CO_2 + water). Ammonium also enters via **potassium channels**.

TREATMENT OF HYPERCHLORAEMIC METABOLIC ACIDOSIS

Agent	Action	Dose	Drawbacks / when to choose
Sodium bicarbonate (oral)	Alkalinises; restores acid-base balance	—	Effective; intestinal gas
Sodium citrate + citric acid (Bicitra, Shohl's solution)	Alkalinises	—	Effective; poor taste
Potassium citrate + sodium citrate + citric acid (Polycitra)	Alkalinises; supplies K^+	—	When sodium load is a problem (cardiac, renal) and potassium is desirable or harmless
Chlorpromazine	Inhibits cyclic AMP → blocks chloride transport	25 mg three times daily (up to 50 mg)	Limits, does not correct acidosis; side effects at higher doses, including tardive dyskinesia
Nicotinic acid	Inhibits cyclic AMP → blocks chloride transport	400 mg three or four times daily	Limits only; avoid in peptic ulcer or hepatic insufficiency ; flushing, dermatitis (adapt), headache, diplopia

- **Chloride-transport blockers** are for persistent acidosis **when sodium loads are undesirable**; they reduce the need for alkali.
- **Hypokalaemia and total-body potassium depletion: commoner with ureterosigmoidostomy** and ureterocolonic diversion than ileal conduits; also in continent diversions; most in **long-standing uncorrected acidosis**.
 - Causes: **renal potassium wasting** (renal damage, osmotic diuresis) and, minor, gut secretion.
 - **Ileum reabsorbs urinary potassium; colon does not** → colon-based diversions deplete more.
 - Severe → **flaccid paralysis**.
 - **Trap**: with severe hyperchloraemic acidosis, **replace potassium as well as bicarbonate** — correcting acidosis alone drives potassium into cells → **severe hypokalaemia and paralysis**.

- **Renal function tests invalidated by the bowel:** **water-deprivation osmolality** (bowel re-equilibrates osmolality); **urine pH** (bowel alkalinises); **serum urea and creatinine** (both reabsorbed by ileum and colon).
- **Chronic urine exposure — histology:** ileum shows patchy **villous atrophy and pseudocrypts** with submucosal inflammation; colon changes fewer (**smaller goblet cells**). Transporters decrease in number but **remain functional** — the **capacity for hyperchloraemic acidosis persists**.

Altered Sensorium

- Causes: **magnesium deficiency** (nutritional or renal wasting, like calcium), **drug intoxication**, **abnormal ammonia metabolism** (commonest). **Hyperglycaemia without glucosuria** occurs in diabetics (urinary glucose reabsorbed) but is not caused by the diversion.
- **Ammoniogenic coma:** in cirrhosis, altered liver function, and even apparently normal liver function; **most commonly with reduced liver function and ureterosigmoidostomy**, also ileal conduits.
- **Treatment:** **drain the diversion** (rectal tube for ureterosigmoidostomy, Foley for continent reservoirs) → **oral neomycin** → **restrict protein** → severe: **intravenous arginine glutamate 50 g in 1000 mL 5% dextrose** (glutamine substrate) → **lactulose** oral or rectal (binds gut ammonia).

Abnormal Drug Absorption

- At risk: drugs **absorbed by the gut and excreted unchanged by the kidney** — re-absorbed from the segment → toxicity. Reported: **phenytoin**, some antibiotics, **methotrexate** (ileal conduit).
- Cisplatin, methotrexate and vinblastine are generally tolerated with normal renal function, though toxicity **trended higher in continent diversions**.
- **Monitor antimetabolite levels; drain continent pouches during chemotherapy.**

Osteomalacia

- **Reduced mineralised bone, excess osteoid** (renal rickets). Reported after colocolostomy, ileal ureter, colon and ileal conduits — **most commonly ureterosigmoidostomy**.

MECHANISMS OF OSTEOMALACIA IN URINARY INTESTINAL DIVERSION

Mechanism	Process	Evidence / remedy
Chronic acidosis	Bone buffers protons, releasing calcium → hypercalciuria	Correcting acidosis remineralises bone; but large bicarbonate falls are not required
Vitamin D resistance (probably renal)	Independent of acidosis	Some do not remineralise on alkali alone → 1α-hydroxycholecalciferol
Renal calcium loss	Reabsorbed urinary sulphate → excreted sulphate inhibits tubular calcium reabsorption	Hypercalciuria and hypermagnesiuria
Parathormone resistance	Altered calcium-to-parathormone ratio long term	Suggestive evidence

- Contributions vary by patient.
- **Prevention:** if a **base deficit >2.5 mEq/L** is corrected, bone mineral density abnormalities do not appear; diversion type matters little once acidosis is corrected. Unselected series show **more acidosis in continent diversions**, and long-term acidotic patients lose bone density. **Correct acidosis meticulously.**
- **Presentation:** **lethargy, weight-bearing joint pain, proximal myopathy**. Biochemistry: **calcium low or normal, alkaline phosphatase raised, phosphate low or normal**.
- **Treatment ladder:** **correct acidosis + calcium** → active vitamin D → **1α-hydroxycholecalciferol**.

Growth and Development

- Long-term follow-up of **myelodysplasia** patients: diversion reduced **linear growth** (significant for **biacromial span and elbow-hand length**).
- Versus ileal-conduit patients, those on **intermittent catheterisation with bladders retained** had fewer **fractures, malunions and non-unions**; more diverted patients fell **below the 10th percentile**, yet **height and weight did not differ**.
- Experimentally, ureterosigmoidostomy shortened femoral length.
- **Diversion in childhood maintained >10 years** → **significant linear-growth changes** on careful study, though not obvious in most.

Infection

- **More bacteriuria, bacteraemia and sepsis; colcystoplasty more than ileocystoplasty**. Acute pyelonephritis occurs in a substantial minority of conduits, and some die of sepsis.
- **About three-quarters of ileal conduit urines contain bacteria**; many are **colonised only distally** (proximal **double-catheter** cultures are far less often positive) and do well.
- **Treat pure cultures of Proteus or Pseudomonas** (predict upper-tract deterioration); **observe mixed cultures if asymptomatic**.
- **Continent diversions**: frequent bacteriuria (about two-thirds of Kock pouches) and sepsis.
- Why: intestine **cannot inhibit bacterial proliferation** (unlike urothelium), may make urine **less bacteriostatic**, and **distension promotes translocation**; mucosal immune defences are largely preserved.

Stones

- Mostly **calcium, magnesium and ammonium phosphate** (infection stones).
- **Highest risk: hyperchloraemic acidosis, pre-existing pyelonephritis, urease-producing infection**.
- **Incidence: colon conduit lowest (3–4%), ileal conduit higher (10–12%), continent caecal reservoirs highest (reservoir calculi ~20%)**.
- **Causes: infection with alkaline urine; hypercalciuria; altered urinary solutes; foreign bodies (staples, non-absorbable sutures) — a major cause; mucus as nidus; mucosal change impairing emptying**.

Intestinal Motility, Short Bowel, and Nutritional Problems

CONSEQUENCES OF LOSING EACH SEGMENT

Segment lost	Consequence
Ileum	Vitamin B12 malabsorption (anaemia, neuropathy); bile-salt malabsorption → colonic irritation, diarrhoea ; loss of the ileal brake (lipid-triggered slowing) → fatty diarrhoea
Ileocaecal valve	Bacterial reflux → small-intestinal bacterial overgrowth ; fat and bile-salt malabsorption → diarrhoea; deficiency of vitamin A, vitamin D (osteomalacia) , calcium lost as soaps; faster transit
Jejunum	Malabsorption of fat, calcium and folate (rarely relevant)
Colon	Diarrhoea (fluid and electrolyte), bicarbonate loss, dehydration

- **Vitamin B12**: low levels are common after ileocystoplasty in children and after irradiated cystectomy, but clinical effects are rarely seen — **liver stores last 3–5 years**, and low serum B12 need not mean deficiency. **Homocysteine and methylmalonic acid** are the sensitive markers (unstudied here). Many give **empirical parenteral B12 after 5–7 years** of ileal diversion.
- Some advocate **reconstructing an ileocolic valve** when the ileocaecal segment is used.

- Removing major segments → **nocturnal bowel actions, urgency, faecal incontinence, diarrhoea**, nutritional deficiency. **Clam cystoplasty** carries more bowel dysfunction than ileal conduit — **detrusor instability is associated with irritable bowel**, so warn such patients.

Cancer

- **Ureterosigmoidostomy: cancer in 6–29% (mean 11%)**, after a **10–20-year latency**. Histology: adenocarcinoma (most), adenomatous polyps, sarcoma, transitional cell carcinoma.
- Case reports after **ileal and colon conduits, augmentations, rectal bladder, neobladders and ileal ureters**.
- **Tissue of origin unresolved — probably both**: most are adenocarcinomas (intestinal); but urothelium exposed to faeces forms adenocarcinoma experimentally, ureters in ureterosigmoidostomy show marked dysplasia, and **removing the urothelium prevents tumours**.
- **Defunctioned ureterosigmoid anastomoses still produce cancer** (reported 22 years after conversion to a conduit) → **excise the old anastomosis, do not merely ligate and leave it**.
- **Highest risk: transitional epithelium juxtaposed to colonic epithelium, both bathed in faeces**.
- Candidate mechanisms: **nitrosamines** (no convincing primary role), abnormal colonic mucin, **ornithine decarboxylase** induction, growth factors (epidermal growth factor) — **cause unknown**.

SECONDARY MALIGNANCY BY DIVERSION TYPE — GERMAN REGISTRY ANALYSIS (KÄLBLE, 2011)

Diversion	Risk	Recommendation
Ureterosigmoidostomy	2.6%	Regular endoscopy from 5 years postoperatively
Cystoplasty	1.6%	
Orthotopic (ileo)colonic neobladder	1.3%	
Catheterisable ileocaecal pouch	0.14%	Evaluate if symptomatic (hydronephrosis, chronic infection, haematuria)
Ileal neobladder	0.05%	No routine surveillance
Ileal conduit	0.02%	

- **Pattern**: risk tracks **colonic mucosa in contact with urine** (and faeces); purely ileal diversions are lowest. The registry's ureterosigmoidostomy figure is lower than the **6–29%** quoted from older ureterosigmoidostomy series — a different denominator (all diversions over 1970–2007, versus long-term ureterosigmoidostomy cohorts).

KEY POINTS

- **Limit bowel length** when the patient **cannot acidify urine below 5.8**, **cannot concentrate above 600 mOsm/kg**, or has **GFR <35 mL/min**.
- Metabolic complications: electrolyte abnormalities, altered sensorium, abnormal drug metabolism, osteomalacia, growth retardation, infection, calculi, short gut, **urothelial or intestinal cancer**.
- **Gastrocystoplasty patients most at risk of severe metabolic alkalosis**: high resting gastrin, pouch overdistension, dehydration.
- **Ileum and colon → hyperchloraemic metabolic acidosis**.
- Chemotherapy toxicity is **increased** with intestine in the urinary tract.

Concept — Nearly every metabolic complication descends from one fact — **bowel absorbs what the kidney excretes** (ammonium chloride, drugs, urea, sulphate, ammonia) — so severity scales with **contact time and surface area**, and the kidney's reserve decides whether the load is tolerated.

SUMMARY

- Complications arise from the diversion itself, its specific type, and the preceding extirpative procedure.
- **Long-term complications significantly contribute to morbidity and mortality**; many cancer patients die of cancer first, so those **diverted for benign disease or cured of cancer** bear the most long-term morbidity.
- Knowledge of complication frequency and **correct preparation, technique and aftercare** gives the best chance of minimal morbidity and mortality.